

Vector-Valued Functions



Position, velocity, and acceleration vectors

- 1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.
 - 2 For the same particle, find the speed at $t = 2$.
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Integrating vector-valued functions

- 3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.
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Total distance traveled

- 4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.
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Vectors and parametric motion combined

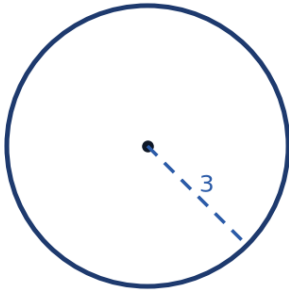
- 5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.
 - 6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.
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Unit tangent vectors

- 7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.
 - 8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.
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Motion on a circle and centripetal acceleration

- 9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).



- 10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t , and interpret this geometrically.

Displacement vs. distance for vector motion

- 11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.
- 12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

Vector operations review

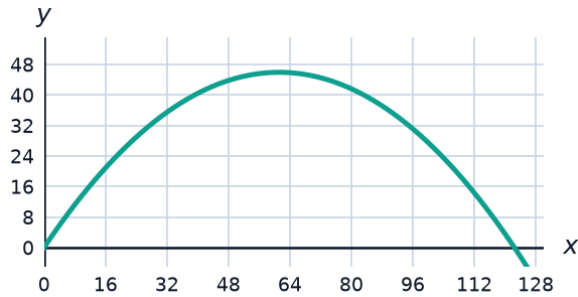
- 13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.
- 14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

Curvature-related speed problems

- 15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.
- 16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ ($t > 0$), find the velocity vector and speed at $t = 1$.

Projectile motion as a vector-valued function

- 17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).



- 18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

Finding when a particle is at rest

- 19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t, t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).
- 20 For the same particle, find the times when only the horizontal velocity component is 0.

Vector-valued functions and arc length

- 21 State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.
- 22 Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.

Multiple choice — easy 15

- 23 Given $\vec{u} = \langle 7, -1 \rangle$ and $\vec{w} = \langle 3, 8 \rangle$, find $2\vec{u} + 3\vec{w}$.
- a. $\langle 23, 22 \rangle$
 - b. $\langle 24, 22 \rangle$
 - c. $\langle 27, 13 \rangle$
 - d. $\langle 23, -22 \rangle$

24 Given $\vec{u} = \langle 6, -8 \rangle$ and $\vec{w} = \langle -1, 6 \rangle$, find $3\vec{u} - 2\vec{w}$.

a. $\langle -15, 34 \rangle$

b. $\langle 20, 36 \rangle$

c. $\langle 21, -36 \rangle$

d. $\langle 20, -36 \rangle$

25 Given $\vec{u} = \langle 7, 7 \rangle$ and $\vec{w} = \langle 2, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.

a. $\langle -7, 10 \rangle$

b. $\langle 17, 5 \rangle$

c. $\langle -8, -10 \rangle$

d. $\langle -8, 10 \rangle$

26 Given $\vec{u} = \langle -5, 0 \rangle$ and $\vec{w} = \langle -8, -4 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle 14, 12 \rangle$

b. $\langle 15, 12 \rangle$

c. $\langle -1, -8 \rangle$

d. $\langle 14, -12 \rangle$

27 Given $\vec{u} = \langle 7, 3 \rangle$ and $\vec{w} = \langle 0, 6 \rangle$, find $2\vec{u} - 2\vec{w}$.

a. $\langle 14, -6 \rangle$

b. $\langle 14, 6 \rangle$

c. $\langle 15, -6 \rangle$

d. $\langle -14, 6 \rangle$

- 28 Given $\vec{u} = \langle -1, 3 \rangle$ and $\vec{w} = \langle 2, 3 \rangle$, find $2\vec{u} - 3\vec{w}$.
- a. $\langle -7, -3 \rangle$
 - b. $\langle -8, 3 \rangle$
 - c. $\langle 7, -3 \rangle$
 - d. $\langle -8, -3 \rangle$
- 29 Given $\vec{u} = \langle -7, -2 \rangle$ and $\vec{w} = \langle 8, -3 \rangle$, find $3\vec{u} - 2\vec{w}$.
- a. $\langle -37, 0 \rangle$
 - b. $\langle -39, 1 \rangle$
 - c. $\langle -36, 0 \rangle$
 - d. $\langle 38, -5 \rangle$
- 30 Given $\vec{u} = \langle 7, -8 \rangle$ and $\vec{w} = \langle -4, 8 \rangle$, find $-2\vec{u} + 3\vec{w}$.
- a. $\langle 29, -40 \rangle$
 - b. $\langle -25, 40 \rangle$
 - c. $\langle -26, -40 \rangle$
 - d. $\langle -26, 40 \rangle$
- 31 Given $\vec{u} = \langle -5, 1 \rangle$ and $\vec{w} = \langle 5, -4 \rangle$, find $2\vec{u} + 3\vec{w}$.
- a. $\langle -5, -5 \rangle$
 - b. $\langle 5, 10 \rangle$
 - c. $\langle 5, -10 \rangle$
 - d. $\langle 6, -10 \rangle$

32 Given $\vec{u} = \langle 6, -7 \rangle$ and $\vec{w} = \langle -7, -7 \rangle$, find $2\vec{u} + 3\vec{w}$.

- a. $\langle -9, 35 \rangle$
- b. $\langle -9, -35 \rangle$
- c. $\langle -8, -35 \rangle$
- d. $\langle 4, -35 \rangle$

33 Given $\vec{u} = \langle 2, 0 \rangle$ and $\vec{w} = \langle 7, -1 \rangle$, find $-2\vec{u} + 3\vec{w}$.

- a. $\langle 17, -3 \rangle$
- b. $\langle 18, -3 \rangle$
- c. $\langle 17, 3 \rangle$
- d. $\langle -8, 2 \rangle$

34 Given $\vec{u} = \langle 1, 3 \rangle$ and $\vec{w} = \langle 5, 4 \rangle$, find $2\vec{u} + 3\vec{w}$.

- a. $\langle 17, -18 \rangle$
- b. $\langle 13, 17 \rangle$
- c. $\langle 18, 18 \rangle$
- d. $\langle 17, 18 \rangle$

35 Given $\vec{u} = \langle 8, -8 \rangle$ and $\vec{w} = \langle -2, -4 \rangle$, find $3\vec{u} + 2\vec{w}$.

- a. $\langle 10, -28 \rangle$
- b. $\langle 21, -32 \rangle$
- c. $\langle 20, 32 \rangle$
- d. $\langle 20, -32 \rangle$

36 Given $\vec{u} = \langle 8, 0 \rangle$ and $\vec{w} = \langle 3, -1 \rangle$, find $2\vec{u} - 3\vec{w}$.

a. $\langle -18, -2 \rangle$

b. $\langle 7, -3 \rangle$

c. $\langle 7, 3 \rangle$

d. $\langle 8, 3 \rangle$

37 Given $\vec{u} = \langle -6, 8 \rangle$ and $\vec{w} = \langle 7, 5 \rangle$, find $2\vec{u} + 3\vec{w}$.

a. $\langle 10, 31 \rangle$

b. $\langle 9, 31 \rangle$

c. $\langle 9, -31 \rangle$

d. $\langle -4, 34 \rangle$

Multiple choice — medium 15

38 A particle moves along $\vec{r}(t) = \langle 5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 26

b. 28

c. 34

d. 13

39 A particle moves along $\vec{r}(t) = \langle 12t^2, -35t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 94

b. 37

c. 74

d. 76

- 40 A particle moves along $\vec{r}(t) = \langle 8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 36
 - b. 17
 - c. 34
 - d. 46
- 41 A particle moves along $\vec{r}(t) = \langle -8t^2, 15t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 36
 - b. 34
 - c. 46
 - d. 17
- 42 A particle moves along $\vec{r}(t) = \langle -4t^2, 3t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 10
 - b. 14
 - c. 12
 - d. 5
- 43 A particle moves along $\vec{r}(t) = \langle -24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 62
 - b. 50
 - c. 52
 - d. 25

- 44 A particle moves along $\vec{r}(t) = \langle 35t^2, -12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 37
 - b. 74
 - c. 76
 - d. 94
- 45 A particle moves along $\vec{r}(t) = \langle -9t^2, -40t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 98
 - b. 84
 - c. 82
 - d. 41
- 46 A particle moves along $\vec{r}(t) = \langle 24t^2, 7t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 62
 - b. 52
 - c. 50
 - d. 25
- 47 A particle moves along $\vec{r}(t) = \langle -8t^2, -6t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 28
 - b. 22
 - c. 20
 - d. 10

- 48 A particle moves along $\vec{r}(t) = \langle 6t^2, -8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 22
 - b. 20
 - c. 28
 - d. 10
- 49 A particle moves along $\vec{r}(t) = \langle 6t^2, 8t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 20
 - b. 10
 - c. 28
 - d. 22
- 50 A particle moves along $\vec{r}(t) = \langle 7t^2, 24t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 62
 - b. 25
 - c. 52
 - d. 50
- 51 A particle moves along $\vec{r}(t) = \langle -8t^2, 6t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 20
 - b. 10
 - c. 28
 - d. 22

- 52 A particle moves along $\vec{r}(t) = \langle -5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.
- a. 13
 - b. 26
 - c. 34
 - d. 28
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Multiple choice — hard 15

- 53 A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.
- a. -2
 - b. 6
 - c. 2
 - d. 3
- 54 A particle moves along $\vec{r}(t) = \langle t^3 - 108t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.
- a. 6
 - b. 18
 - c. 7
 - d. -6

- 55 A particle moves along $\vec{r}(t) = \langle t^3 - 48t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.
- a. 4
 - b. 5
 - c. 12
 - d. -4
- 56 A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 - 6 \rangle$. Find the positive time t when the horizontal velocity component is 0.
- a. 2
 - b. 3
 - c. 6
 - d. -2
- 57 A particle moves along $\vec{r}(t) = \langle t^3 - 75t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.
- a. 5
 - b. 6
 - c. -5
 - d. 15
- 58 A particle moves along $\vec{r}(t) = \langle 7t, 48t - 4t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).
- a. 7
 - b. 6
 - c. 12
 - d. 5

- 59 A particle moves along $\vec{r}(t) = \langle 4t, 6t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).
- a. 1
 - b. 2
 - c. 3
 - d. 4
- 60 A particle moves along $\vec{r}(t) = \langle 4t, 20t - 2t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).
- a. 10
 - b. 6
 - c. 5
 - d. 4
- 61 A particle moves along $\vec{r}(t) = \langle 6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).
- a. 12
 - b. 7
 - c. 5
 - d. 6
- 62 A particle moves along $\vec{r}(t) = \langle -6t, 36t - 3t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).
- a. 7
 - b. 12
 - c. 5
 - d. 6

63 Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 2]$.

- a. 82
- b. 98
- c. 123
- d. 84

64 Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 7]$.

- a. 119
- b. 136
- c. 161
- d. 126

65 Find the arc length of $\vec{r}(t) = \langle 20t, 21t \rangle$ for $t \in [0, 4]$.

- a. 164
- b. 145
- c. 120
- d. 116

66 Find the arc length of $\vec{r}(t) = \langle 9t, 40t \rangle$ for $t \in [0, 4]$.

- a. 164
- b. 205
- c. 196
- d. 168

67 Find the arc length of $\vec{r}(t) = \langle 8t, 15t \rangle$ for $t \in [0, 3]$.

a. 68

b. 54

c. 69

d. 51